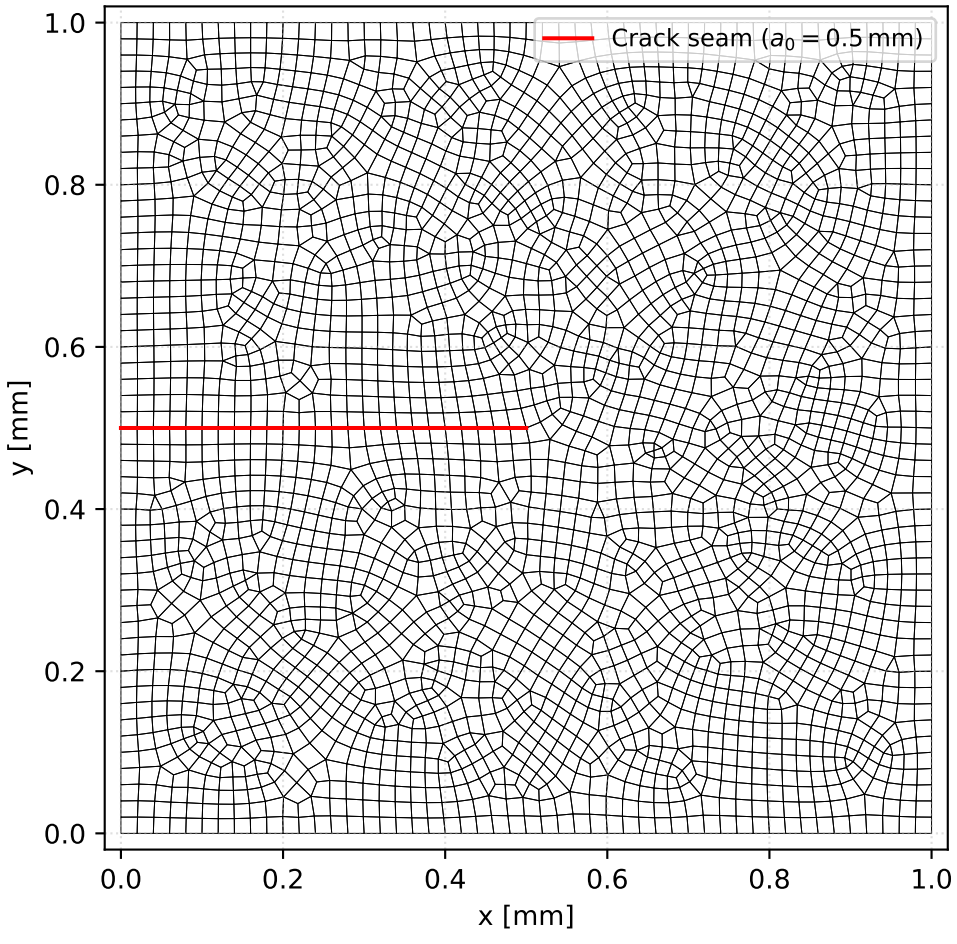
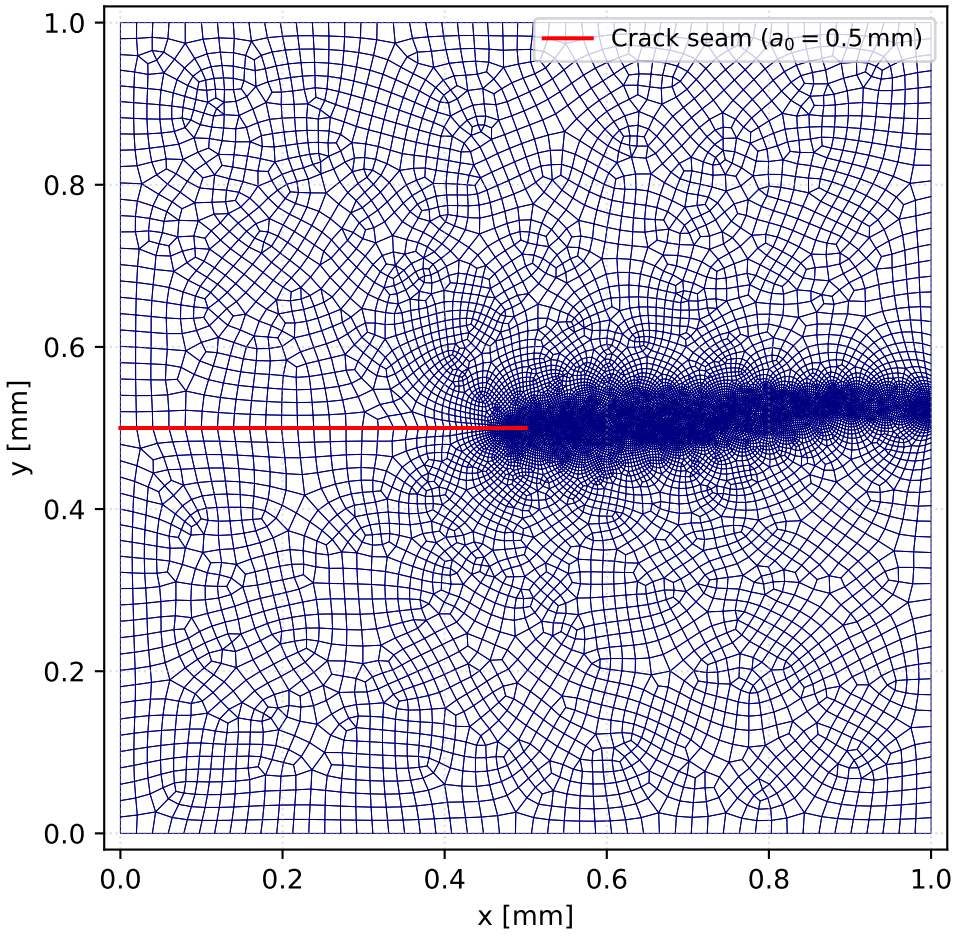
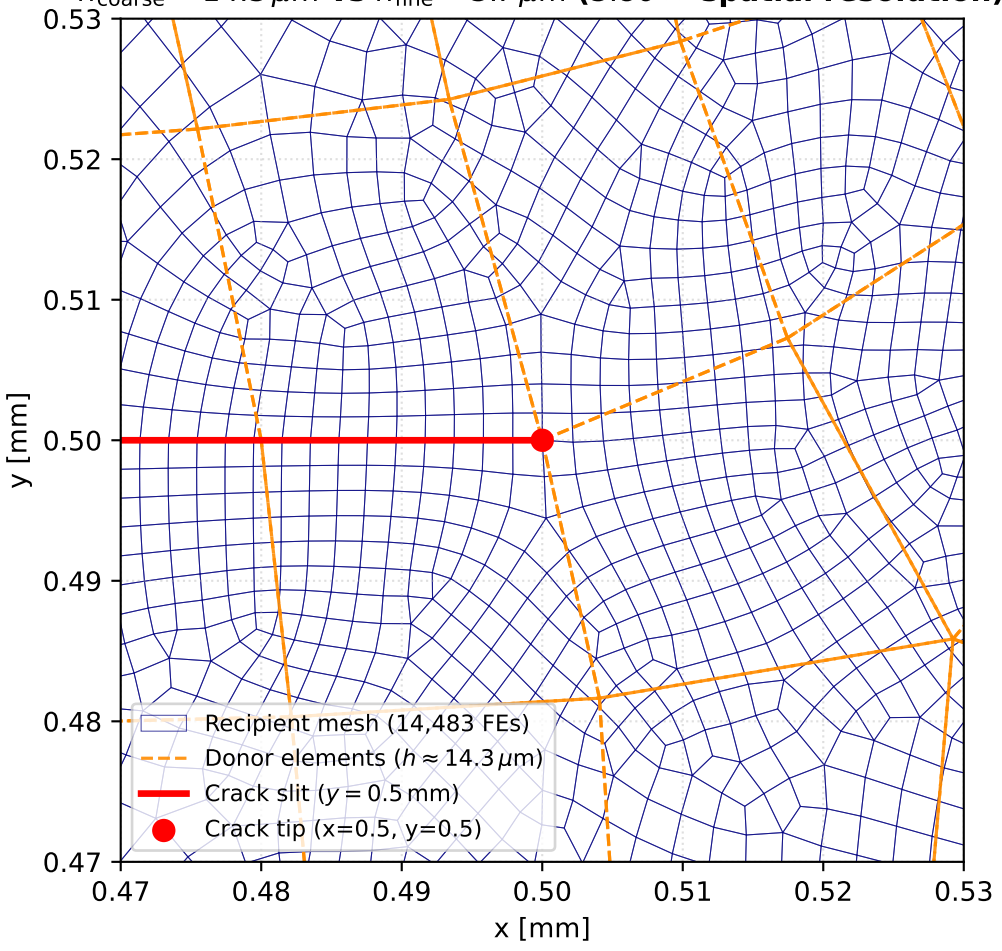
(a) Coarse Donor Mesh: 2,906 Finite Elements
Job 1413814 (Package 48), $h \approx 14.3\,\mu$m **corridor**



(b) Refined Recipient Mesh: 14,483 Finite Elements
Job 1412465 / P49 Target, $h \approx 3.7\,\mu$m **corridor**



(c) Crack-Tip Discretization Contrast (Close-Up)
$h_{\text{coarse}} \approx 14.3\,\mu$m **vs** $h_{\text{fine}} \approx 3.7\,\mu$m **(3.86 × spatial resolution)**



(d) Matched-Displacement Phase-Field Damage Profiles
at Handoff State $u = 0.005000$ mm **along Ligament (**$y = 0.5$ mm**)**

